

Unit 11, Lesson 7: Translating Real-World Written Descriptions into Algebraic Inequalities

Benchmark

MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

Learning Targets

- ☐ I can translate a written description into an algebraic inequality.
- ☐ I can represent an inequality on a number line when given a written description of a real-world context.

11.7.1 Warm-Up: Jumbotron Estimation

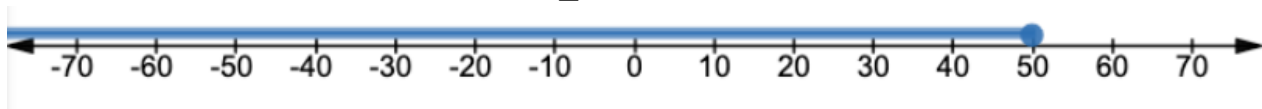
In the U.S., the average LCD TV screen has a width of 3.63 feet (ft) and a height of 2.04 ft.

Doak Campbell stadium has a jumbotron screen with a width of 119.7 ft and a height of 56.7 ft.

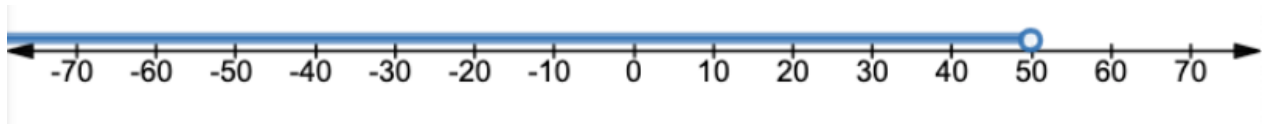
1. Estimate how many TV screens can be laid on top of the jumbotron at Doak Campbell stadium.
2. Explain how you found your estimate.

11.7.2 Exploration: Representing Inequalities on a Number Line

1. Complete the following problems.
 - a. Choose an x -value that will make the inequality $x \leq 50$ true.
 - b. Choose an x -value that will make the inequality $x \leq 50$ false.
 - c. Have your partner verify your choices are correct.
 - d. What is the x -value that is true for $x \leq 50$ but false for $x < 50$?
 - e. The number line below represents all of the x -values that are true for $x \leq 50$.



The number line below represents all of the x -values that are true for $x < 50$.

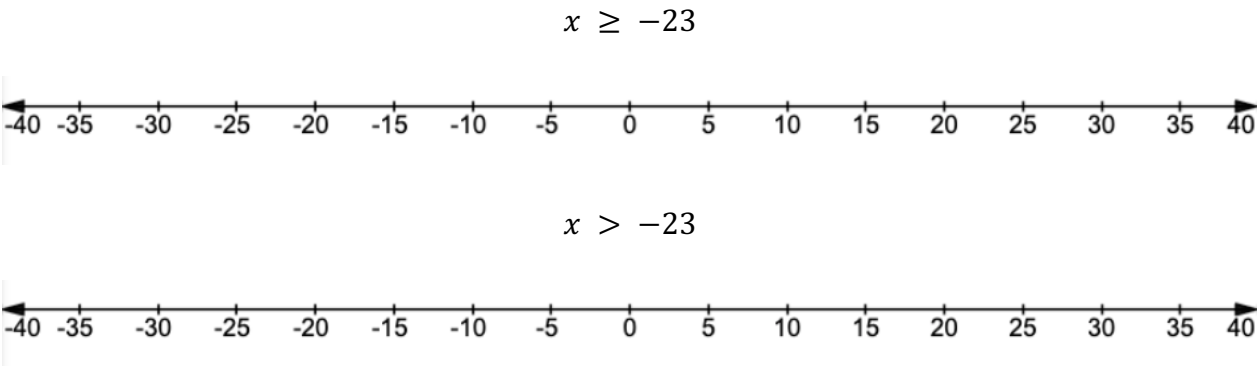


How is the graph for $x < 50$ different from the graph for $x \leq 50$?

2. Complete the following.
- a. How are the solutions to $x \geq -23$ and $x > -23$ the same?

b. How are the solutions to $x \geq -23$ and $x > -23$ different?

c. Graph the solutions to each inequality on the number lines.



3. Complete the following.
- a. With your partner, determine which inequality best represents each context.

Nutritional Information		
	Calories:	2,000
Total Fat	Less than	65g
Sat Fat	Less than	20g
Cholesterol	Less than	300mg
Sodium	Less than	2,400mg

Total Fat

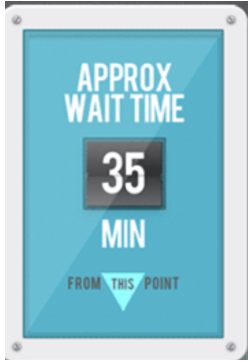
$f \leq 65$ $f \geq 65$

$f < 65$ $f > 65$

YOU MUST BE AT LEAST
54"
TALL TO RIDE

$h \leq 54$ $h \geq 54$

$h < 54$ $h > 54$

 $p \leq 10$ $p \geq 10$ $p < 10$ $p > 10$	 $s \leq 50$ $s \geq 50$ $s < 50$ $s > 50$
 $x \leq 50$ $x \geq 50$ $x < 50$ $x > 50$	 $w \leq 35$ $w \geq 35$ $w < 35$ $w > 35$

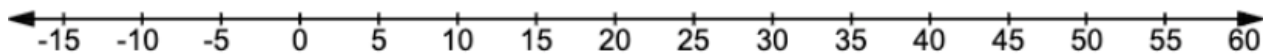
- b. In which, if any, of these situations does a value of 0 make sense given the context?
- c. In which, if any, of these situations does a negative number make sense given the context?
- d. In which, if any, of these situations does 1,000 make sense given the context?

11.7.3 Guided Practice: Translating a Written Description into an Algebraic Inequality

1. According to the Interstate Highway Standards, U.S. and state highway traffic lanes must be at least 15 feet wide.

a. Write an inequality to represent the width, w , that traffic lanes can be.

b. Represent the inequality on the number line.



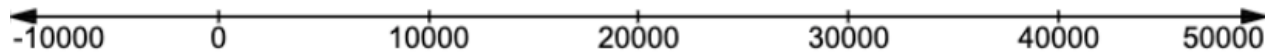
c. List two examples of possible widths, in feet, for the traffic lane.

d. Give one example of a width that is true for the inequality but does not make sense in the real world.

2. High-altitude parachute jumps are usually made for an altitude that is no more than 35,000 feet to avoid frostbite from extreme cold.

a. If h represents the height of a jumper, write an inequality to describe all possible values of h .

- b. Represent the inequality on the number line.

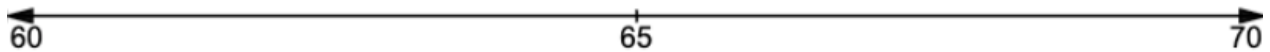


- c. Does the inequality that represents the situation include 0 as a possible height?
- d. Explain why a negative number does not make sense for this context.

11.7.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

1. The inequality $67.85 \leq m$ represents the amount of money, m , Edin needs to earn to buy a new bike.

- a. Graph the inequality.



- b. Give one value that is true for $67.85 \leq m$ but is unreasonable for the context.

- c. Explain why the value you chose is unreasonable.

2. A farmer plans to plant more than 52.75 acres of oranges.

- a. If A represents the number of acres of oranges the farmer plants, write an inequality that represents the number of acres the farmer plants.

- b. Circle all the possible values for A , the number of acres the farmer plans to plant, from the list below.

52.7 acres

$52\frac{3}{4}$ acres

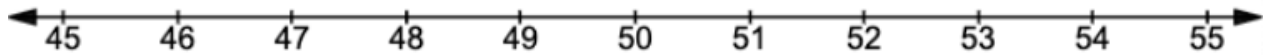
53 acres

52 acres

52.8 acres

$52\frac{1}{4}$ acres

- c. Graph the inequality.



11.7.5 Your Turn!

1. Would you rather drive a go-cart with speed, g , in miles per hour (mph), represented by the inequality $12.5 \geq g$, or an electric bike with speed, b , in mph, represented by $b \geq 7\frac{1}{2}$?

a. I choose the

- ☐ go-cart
 - ☐ electric bike

because _____

_____.

The

- ☐ go-cart
 - ☐ electric bike

could have a speed of 18 mph.

- b. Circle all the values from the list below that are possible speeds for the go-cart.

1.25 mph

15.2 mph

$10\frac{1}{3}$ mph

$\frac{12}{5}$ mph

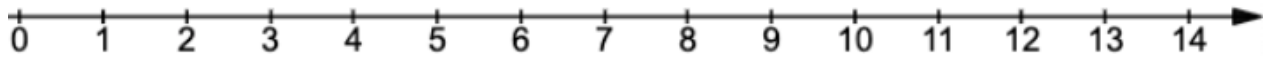
12.45 mph

11.95 mph

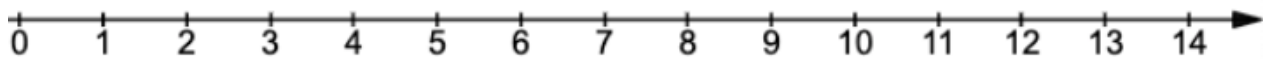
$12\frac{2}{3}$ mph

$11\frac{11}{12}$ mph

- c. Represent the speed of the electric bike on the number line.



- d. Represent the speed of the go-cart on the number line.



- e. Explain why the number lines do not include negative numbers.

11.7.6 Wrap-Up: Ticket Out the Door

At Universal Studios, a person must be at least 48 inches (in.) tall to ride Revenge of the Mummy™.

1. Write an inequality representing the height, h , a person must be to go on the ride.
2. Circle all values from the list below that satisfy this inequality.

47.9 in.

100 in.

 $\frac{48}{2}$ in.

48.5 in.

 $70\frac{3}{4}$ in. $48\frac{1}{4}$ in.

0 in.

400 in.

48.099 in.

48 in.

 $50\frac{3}{4}$ in.

47.9 in.

3. Graph the inequality on the number line.

